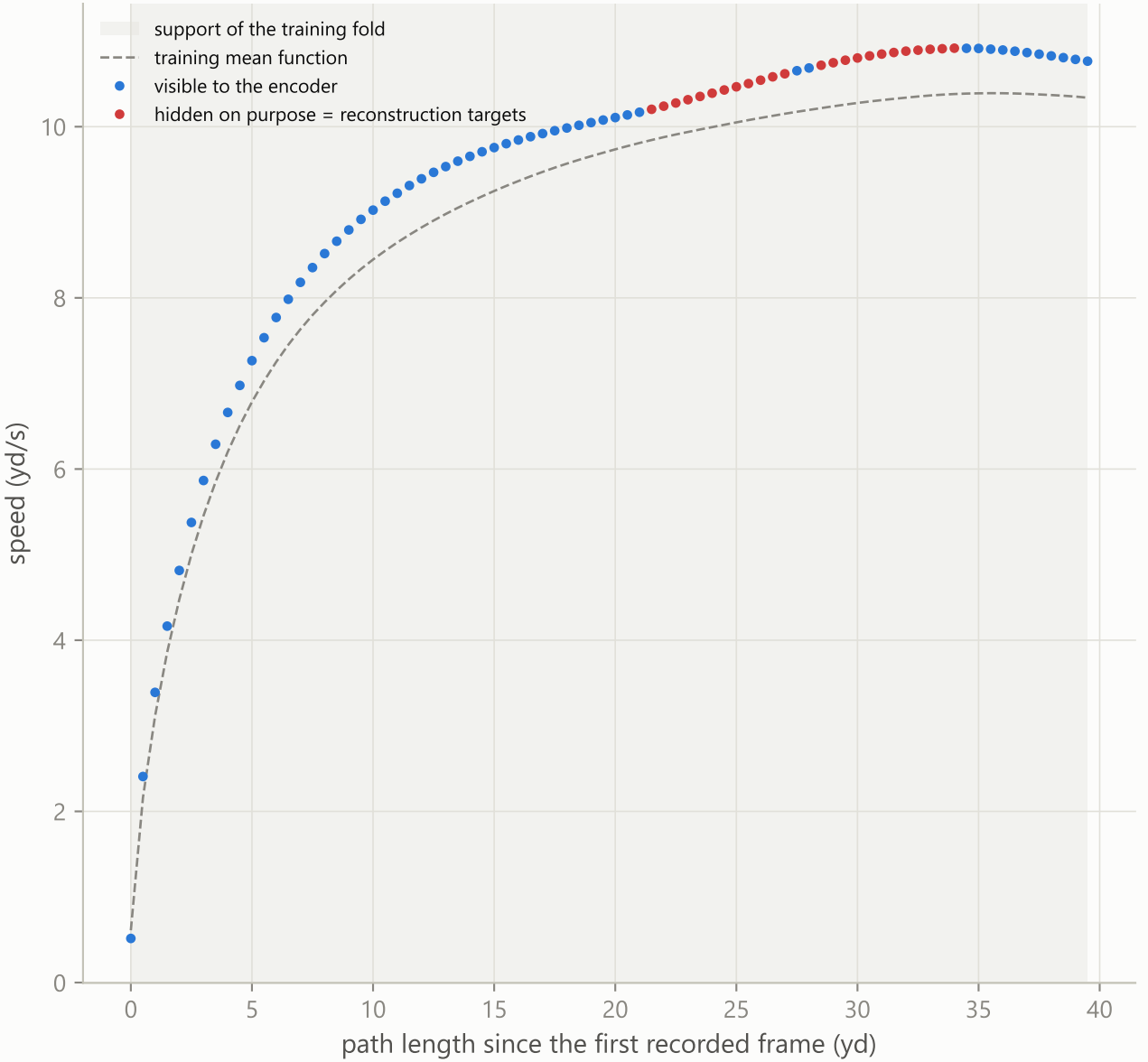
Input representation and encoder

Input: one recorded attempt and its masks



Encoder: 32,761 trainable weights · receptive field 57 grid points

3 channels per grid point

value (0 where not visible) · visible flag · grid position

↓

residual block, dilation 1

2 × convolution (kernel 5, 32 channels), GELU, dropout 0.1

↓

residual block, dilation 2

2 × convolution (kernel 5, 32 channels), GELU, dropout 0.1

↓

residual block, dilation 4

2 × convolution (kernel 5, 32 channels), GELU, dropout 0.1

↓

pooling over VISIBLE points

mean and max; padding, unrecorded and hidden points never enter

↓

embedding z · 8 numbers

the only thing the decoder and every evaluation see

↓

decoder

z → 64 → the curve on 81 grid points; loss on hidden points only

Left: one real attempt of a held-out athlete, A · speed against recorded distance. Values are never filled: a hidden or unrecorded point enters the encoder as 0 with the visible flag at 0. Right: the model. TCN-B adds a consistency term between the embeddings of two attempts of the same TRAINING athlete, encoded on the grid points both recordings hold.